

HAT-P-12b

R-1492

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_210608 · created 2026-08-01T13:58:22+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459373.7279 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.018 +/- 0.053
Transit depth (Rp/Rs)^2	0.031 +/- 0.19 [%]
Orbital Inclination (inc)	84.28 +/- 1.72
Airmass coefficient 1 (a1)	0.4 +/- 0.32
Airmass coefficient 2 (a2)	0.8 +/- 1.01
Scatter in the residuals of the lightcurve fit is	91.46 %
Best Comparison Star	#3 - [235, 249]
Optimal Aperture	1.5
Optimal Annulus	14.49
Transit Duration (day)	0.018 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.018 ± 0.053 vs 0.1406 (deviation 1.36σ)
Duration fit vs published	0.018 d vs 0.0975 d (deviation 1.73σ)
Mid-time O–C	-5.1 min
Depth SNR	0.2
Residual scatter	91.46 %

Light curve

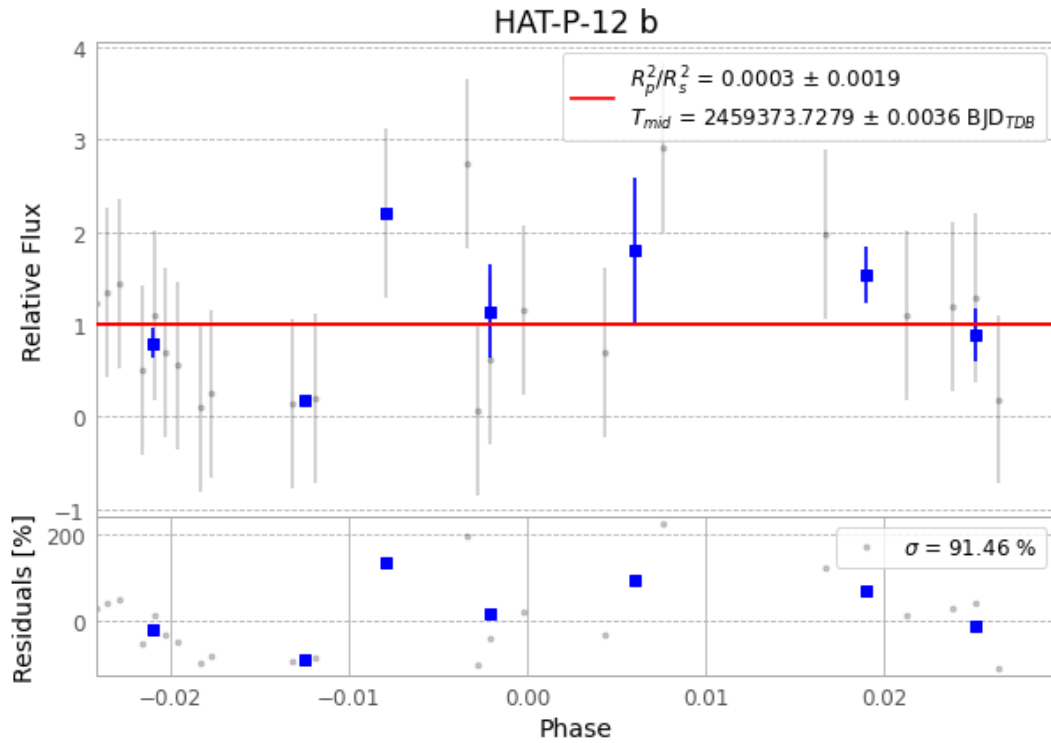


Figure 1 — FinalLightCurve_HAT-P-12 b_2021-06-07.png (SHA-256 b0daf5ac0442716f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 209], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 28b156e5a9a71860...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

86 FITS frames (56 MB), HATP-12210608033312.FITS → HATP-12210608074812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210608.log (7377ed9872b4...), FinalLightCurve_HAT-P-12 b_2021-06-07.png (b0daf5ac0442...), FinalLightCurve_HAT-P-12 b_2021-06-07.csv (393c66d2afeb...)

Compute resources

Wall time 310.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 13:58Z.